Rigorous Parallax Observation Model-Based Remote Sensing Panchromatic and Multispectral Images Jitter Distortion Correction for Time Delay Integration Cameras

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***Abstract*— Time delay integration charge-coupled device (TDI CCD) is sensitive to the platform's stability during push-broom imaging. Due to variations in total integration time, panchromatic and multispectral images suffer varying degrees of geometric distortion caused by satellite jitter with high frequency, which leads to different inner distortion in different band images and different band-to-band mismatching errors between different band combinations. To address this problem, this article proposes a rigorous parallax observation model considering multistage integration time and presents a jitter distortion correction method for remote sensing panchromatic and multispectral images captured by TDI cameras based on it. First, the law of the amplitude attenuation and phase offset of platform jitter deviation on the image under different TDI stages is determined through simulation verification. Then, the rigorous parallax observation model is proposed to establish an accurate relationship between the relative jitter error of two multispectral images with multistage integration and the absolute single-stage integration jitter error by introducing the amplitude attenuation factor and phase offset. Finally, the jitter distortion curves of images with different integration stages and integration time can be reconstructed based on the estimated absolute jitter error and the imaging parameters. Subsequently, the jitter distortion can be further corrected by image resampling. The proposed method was verified through both simulation and real data experiments using GaoFen-9 satellite images. Experimental results show that the proposed method can effectively correct high-frequency jitter distortion in panchromatic and multispectral images, which cannot be corrected by traditional single-stage integration jitter detection model.**

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***Index Terms*— Distortion correction, panchromatic and multispectral images, rigorous parallax observation model, satellite jitter, time delay integration charge-coupled device (TDI CCD).**

I. Introduction

DURING the on-orbit operation period of high-resolution optical satellites, there is an attitude vibration of the satellite platform with small amplitude and high frequency [1], [2]. This vibration is caused by factors such as significant temperature variations due to orbital changes or the periodic movement of moving parts carried on the platform [3], [4]. This kind of platform jitter is a common phenomenon that cannot be overlooked. It can result in geometric distortion of imagery from high-resolution satellites such as QuickBird [5], Advanced Land Observing Satellite (ALOS) [6], Pleiades [7], Ziyuan-1-02C (ZY1-02C) [8], and Ziyuan-3 (ZY-3) [9], [10].

Time delay integration charge-coupled device (TDI CCD) is a kind of linear CCD that uses time delay integration technology. It uses synchronous accumulation with multistage integration time to extend the exposure time [11], [12]. This greatly reduces the weight, volume, and cost of the camera system [13] and achieves high sensitivity and imaging efficiency without sacrificing spatial resolution [11]. Therefore, TDI CCD cameras are widely used in high-resolution remote sensing satellites [1], [14].

Fig. 1(a) displays the ideal working process of TDI CCD, where M is the pixel number of each row, and N is the integration stages of TDI CCD. Once the first pixel in the first row of TDI CCD integrates the target object, the generated charge is quickly transferred to the second pixel in the subsequent row in synchronization with the optically scanned image. This process continues until N integration periods are completed, at which point the accumulated charge is transferred to the horizontal shift register and the image of that row is read out [15], [16]. Therefore, it is necessary to ensure that the charge transfer speed of the TDI CCD is synchronized with the moving speed of the target object. During satellite operation, the slight jitter of satellite platform can lead to varying imaging offsets at each stage. This can cause continuous dislocation and aliasing of charges in regions, ultimately resulting in image blurring and

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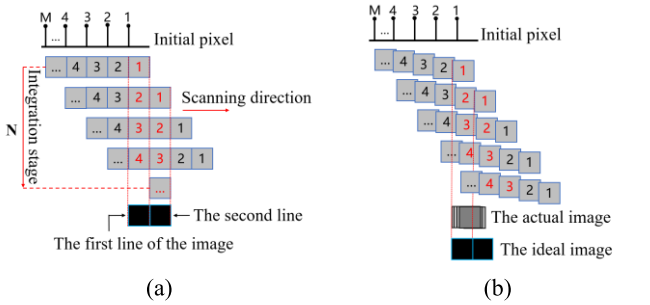

Fig. 1. Integral imaging process of the TDI CCD. (a) Ideal situation without jitter. (b) Actual situation with jitter.

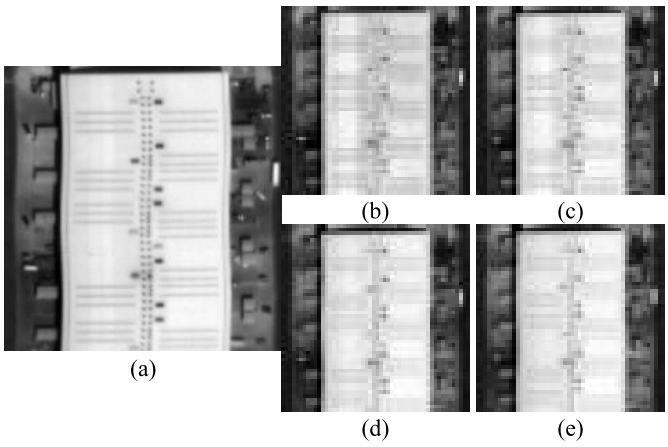

Fig. 2. Panchromatic and multispectral images from the GaoFen-9 satellite of the same area. (a) Panchromatic image. (b)–(e) B1, B2, B3, and B4 bands of multispectral images.

distortion [16], [17], as shown in Fig. 1(b). As the integration stages of TDI CCD increase, the impact of jitter offsets on image quality becomes more pronounced. In comparison to single-line linear CCD, TDI CCD with multistage integration is more sensitive to satellite jitter.

Most of the existing high-resolution push-broom space remote sensing cameras often use TDI CCD as detectors, which contain panchromatic and multispectral bands, including the blue, green, red, and near-infrared bands [2]. The panchromatic and multispectral bands are arranged in parallel at certain intervals on the same focal plane of TDI CCD cameras. Generally, the spatial resolution of panchromatic image is four times that of multispectral image. The single integration time of multispectral image is four times longer than that of panchromatic image.

Fig. 2 shows the same area in panchromatic and multispectral images of the GaoFen-9 satellite. The panchromatic image has single integration time of 0.0906 ms, and the integration stage of 48 TDI stages. Multispectral images have single integration time of 0.3624 ms, and the integration stages of B1, B2, B3, and B4 band images are 48, 48, 24, and 24 TDI stages, respectively. Compared with the multispectral image, the panchromatic image exhibits more obvious jitter distortion because of the smaller ground sampling distance. The integration stage of B1 and B2 band images is greater than that of B3 and B4 band images, indicating a greater jitter attenuation in B1 and B2 bands. B1 and B2 bands display minimal jitter distortion, while B3 and B4 bands exhibit more notable distortion. It is necessary to accurately model the absolute jitter error of each band image to correct the jitter distortion of panchromatic and multispectral images for high-precision applications.

To address the problem, we propose a rigorous parallax observation model-based remote sensing panchromatic and multispectral images jitter distortion correction method for TDI cameras. The primary innovation of this article lies in the following.

1) Through theoretical analysis and simulation verification, the law of the amplitude attenuation and phase offset of the platform jitter deviation on the image under different TDI stages was determined. The amplitude of the jitter error follows the attenuation law of the absolute value of $\sin c$ and the initial phase changes periodically with the total integration time.
2) By introducing the jitter distortion amplitude attenuation factor and phase offset, the rigorous parallax observation model was proposed to establish the relationship between the relative jitter error of two multispectral images with multistage integration and the absolute single-stage integration jitter error.
3) Based on the model, a jitter distortion correction method for panchromatic and multispectral images with multistage integration was proposed. The absolute jitter error of imaging under different integration stages and integration time can be obtained according to the estimated absolute single-stage integration jitter error. So, the jitter distortion can be corrected by image sampling. Furthermore, the experimental results show that the proposed method outperforms the traditional method for jitter modeling with high frequency.

II. Related Works

The impact of platform jitter on the imaging quality of TDI CCD has been a topic of interest for researchers. Both single-stage CCD and multistage TDI CCD require platform jitter detection before subsequent applications such as geometric positioning and change detection of the image. Platform jitter detection methods can be divided into direct detection and indirect detection. One commonly used indirect detection method is based on remote sensing images. It mainly uses the characteristics of optical satellite parallax imaging [18] (including multispectral images, noncommon-line CCD images [19], stereo images [20], etc.), ortho-images [21], [22], sequence CMOS images [23], [24], and image edge features [25] to obtain platform jitter data.

The traditional single-stage CCD jitter detection is often based on the characteristics of parallax imaging. Parallax imaging broadly refers to the nontime synchronous imaging of multiple CCDs within an imaging payload or multiple imaging payloads on the same satellite platform [26]. The design of parallax imaging directly leads to imaging parallax between parallax imaging imagery. There is a difference in coordinate between the positions of the same ground object in different CCD imaging images, both across-track and along-track.

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For images of parallax imaging with short time intervals, such as multispectral images, the small distance between different bands causes them to capture the same scene at different times. The parallax is a constant value. In the presence of platform jitter, the disparity between parallax imaging images is no longer a constant value, but is a periodic variable that changes over time. Therefore, it is possible to detect the disparity in the across-track or along-track direction caused by platform jitter based on the overlapping area between parallax imaging images. It can be expressed by the following equation [27]:

$$g(t) = d(t + \Delta t) - d(t) \tag{1}$$

where t is the current imaging time, and Δt is the imaging time interval. $d(t)$ is the absolute distortion caused by jitter with single-stage integration, also called the jitter image offset. $g(t)$ is the relative error between multispectral images caused by jitter with single-stage integration.

Tong et al. [27] first proposed the framework for jitter detection and compensation using images. The comprehensive image matching approach was used to detect the jitter around 0.67 Hz in the across-track direction of ZY-3 based on three-line array stereo images [28]. In addition, the subpixel phase correlation image matching method was proposed to detect the jitter at 0.65 Hz of ZY-3 using multispectral images [29]. Zhu et al. [1] also detected the platform jitter of the ZY-3 satellite with a frequency of 0.6 Hz by least-squares dense matching using multispectral imagery. Sun et al. [30] detected three different frequency jitters of 0.1, 0.635, and 4 Hz in the along-track direction of the mapping satellite-1, which adopted the jitter detection method based on short-time asynchronous satellite images.

In 2016, Wang et al. [31] developed the jitter detection and compensation model based on multispectral images. They improved the registration accuracy of multispectral images from the ZY-3 satellite. Subsequently, Pan et al. [32] conducted further research and established an accurate jitter displacement estimation model based on single-stage CCD images for the first time. ZY-3 and ZY1-02 satellite data were used for experimental validation. The results showed that platform jitter could be accurately estimated, and the model could be applied to parallax images at any imaging interval. Zhu et al. [33] and [34] investigated jitter detection of multispectral imagery for the GF-1 02/03/04 satellite. They first detected satellite jitter with a frequency of 1.1–1.2 Hz in the cross-track direction. Hu et al. [2] proposed a high-frequency jitter detection method for satellite platforms and found that the China Mapping Satellite-1 exhibited high-frequency jitter. Liu et al. [35] suggested a technique that integrates attitude sensors with attitude data from parallax observations to establish a time-domain jitter detection model. They could detect the global jitter, including initial jitter, and found that the Chinese Heavenly Palace-1 satellite has a jitter of 0.12 Hz in the cross-track direction.

However, the above research has more often approximated the TDI CCD as a single-stage integration CCD for jitter detection. Most of the detected jitter is low frequency, which has not been sufficiently sensitive to the TDI CCD integration stages, making the effect easily ignorable. However, as satellite resolution has advanced, the influence of high-frequency jitter on multistage integration images has become increasingly significant and cannot be overlooked.

Nowadays, there are relatively few studies on jitter distortion analysis and modeling of multistage TDI CCD images, which are not yet mature.

In early times, Zhuang [15] and Wang [16] found that low-frequency jitter can cause spatially varying blurring and geometric distortion. Increasing the number of TDI CCD integration stages can reduce the degree of image distortion. The change in the initial jitter phase will result in the deformation of the image to phase shift in the spatial domain. But these studies remain in their qualitative relationship, and there is no further quantitative analysis. By analyzing the deformation law of multistage integration TDI CCD image caused by platform jitter, Zhu [26] first established the platform jitter detection model of multistage integration image. The numerical simulation and image simulation experiments were conducted using the image motion formula, following the principles of photogrammetry. The experimental results show the feasibility of the jitter model. However, it focuses on jitter detection and does not address the analysis of jitter distortion characteristics for TDI CCD images with multistage integration.

In recent years, Ye et al. [36] first proposed the integration imaging model for jitter detection. Subsequently, Pan et al. [17] proposed a continue dynamic shooting model (CDSM) to reflect the real push-broom mode, along with an integral transformation function to improve jitter detection performance. They increased the accuracy of jitter detection and improved the geometric and radiometric quality of multispectral images. But they did not correct the jitter distortion of the panchromatic image. Through theoretical derivation, Liu et al. [37] established the accumulated jitter formula after TDI integration and proposed a model for estimating jitter amplitude of TDI CCD images with attenuation compensation. However, they did not validate the model using real image data. Li et al. [38] conducted blind restoration research using deep learning for TDI CCD images through a fast imaging simulation method, which can effectively restore degraded images. But they did not analyze the geometric distortion of the image affected by jitter from the perspective of degradation mechanisms.

III. Methodology

The jitter distortion correction method for high-resolution remote sensing panchromatic and multispectral images is proposed based on the rigorous parallax observation model for TDI cameras. The workflow shown in Fig. 3 mainly includes four steps. First, the jitter attenuation law of TDI CCD images with multistage integration is analyzed to determine the amplitude attenuation factor and phase offset for various integration stages and integration time. Second, the relative jitter error with multistage integration is determined through the parallax map of two multispectral images by dense matching. Then, the absolute jitter error with single-stage integration for multispectral images is inversed based on the rigorous parallax observation model, which establishes the

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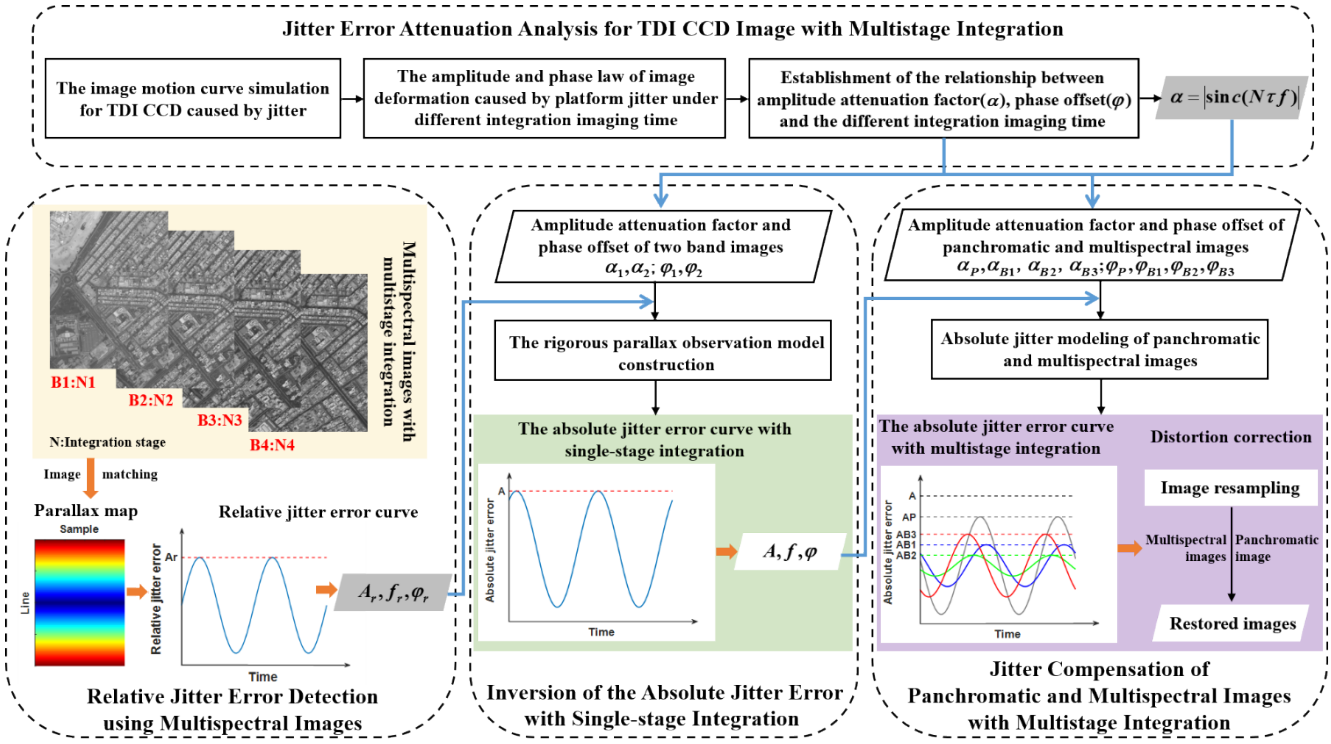

Fig. 3. Workflow of the rigorous parallax observation model-based remote sensing panchromatic and multispectral images' jitter distortion correction method for TDI cameras.

relationship between the relative jitter error of two multispectral images with multistage integration and the absolute jitter error with single-stage integration. Finally, the jitter compensation for panchromatic and multispectral images with multistage integration is carried out. According to the law of jitter amplitude attenuation and phase offset, the absolute jitter error of panchromatic and multispectral images at various integration stages and integration time can be determined using the estimated absolute single-stage integration jitter error. So, the jitter distortion can be further corrected by image sampling for panchromatic and multispectral images, respectively.

A. Jitter Error Attenuation Analysis for TDI CCD Image With Multistage Integration

The final output image line of TDI CCD images with multistage integration is the synthesis of a multistage integration image. Therefore, the jitter on images is the synthesized effect of the image at each integration stage. Considering that the image motion caused by platform jitter varies with time, and it is influenced by the image at each integration stage during the multistage integration imaging process of TDI CCD. The synthesized image is no longer affected in the same way as that of single-stage integration image.

The jitter curve can be described by the sine function, i.e., $\varepsilon(t) = A\sin(2\pi t/T + \varphi)$. A and φ represent the amplitude and initial phase of image motion caused by platform jitter under a single integration time, respectively. T is the period of the jitter curve. The imaging time of each row is divided into N equal parts, where N corresponds to the TDI CCD integration stages. Using the integration time of the first stage as the imaging time for multistage integration images, the image motion with multistage integration affected by jitter can be expressed as follows [17]:

$$\begin{aligned} d_{\mathrm{TDI}}(i, M) &= \frac{1}{MN}\left[\frac{1}{2}A\sin\left((i+N)\frac{2\pi\tau}{T}+\varphi\right)\right. \\ &\quad + \sum_{k=1}^{MN-1} A\sin\left(\left(i+N+\frac{k}{M}\right)\frac{2\pi\tau}{T}+\varphi\right) \\ &\quad \left. + \frac{1}{2}A\sin\left(i\cdot\frac{2\pi\tau}{T}+\varphi\right)\right] \end{aligned} \tag{2}$$

where i is the image row, τ is the single integration time of the TDI CCD, and M is the subdivision numbers of the CDSM [17] (the value of M is 10 in this article). CDSM provides a denser sampling strategy to reduce associated error. It divides imaging time of a single-row into $N * M$ parts, resulting in a total of $(N * M + 1)$ sampling points.

According to (2), we simulated the image motion curves caused by platform jitter at various jitter frequencies and integration stages. Subsequently, sinusoidal curve fitting was performed on the image displacement curve to obtain the amplitude and phase of image deformation under various frequencies and integration stages. According to the characteristics of TDI CCD, it is known that the increasing number of integration stages implies the extension of integration time. We know the impact of platform jitter on images at various integration stages is related to the integration time and jitter frequency. Therefore, we furtherly obtained the relationship between integration time, amplitude attenuation factor, and phase offset.

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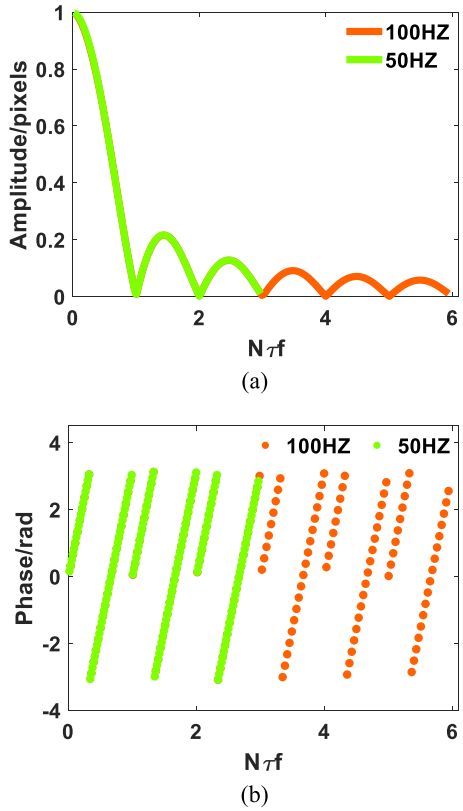

Fig. 4. Jitter attenuation law for TDI CCD images under different integration time and frequency. (a) Amplitude attenuation curves at various integration times and frequencies. (b) Phase offset curves at various integration times and frequencies.

We conducted the simulation experiments using (2). To facilitate the analysis, given the jitter amplitude of 1 pixel (the maximum deformation for an image with single-stage integration is 1 pixel), the jitter initial phase of 0 rad, and the single integration time (τ) of the camera is 0.00036 s. Under the condition that the platform has jitter frequencies of 50 and 100 Hz, we simulated the TDI CCD image motion curve caused by platform jitter from a single stage ($N = 1$) to 167 integration stages, where $T = (1/f), i = 5000$. We then analyzed the law of amplitude attenuation and phase offset of jitter error on multistage integration image.

We obtained the relationship curve between the product of integration stages, single integration time, platform jitter frequency ($N\tau f$), and amplitude and phase, as shown in Fig. 4. The experimental jitter amplitude is 1 pixel, and the initial phase is 0 rad. So, the amplitude and phase of the image deformation are also referred to as the amplitude attenuation factor and phase offset. The amplitude attenuation factor and phase offset can be obtained based on the value of $N\tau f$ when the integration stage, single integration time, and jitter frequency are all determined.

Fig. 4(a) shows that as the TDI CCD integration stages increase, the image deformation amplitude (amplitude attenuation factor) gradually decreases caused by jitter when the single integration time and jitter frequency are fixed. It means that the degree of image deformation is gradually reduced, and the degree of jitter attenuation is increased. Similarly, it can be seen from Fig. 4(b) that with the increasing number of integration stages, the image exhibits a phase offset. The speed of phase offset also varies with platform jitter frequency, and the phase offset caused by high-frequency jitter is periodic.

When $N\tau f$ is an integer, $N\tau f = N\tau(1/T) = 1, 2, 3, \ldots,$ The imaging time $N\tau$ is exactly a multiple of the jitter period. During a jitter period, the image motion of the multistage integrated image is zero, and both the amplitude and phase of image deformation are also zero. After the initial jitter period, when the imaging time is half of the jitter period, $N\tau f = N\tau(1/T) = 1.5, 2.5, 3.5, \ldots$ The image motion of multistage integral images reaches its the maximum value during the jitter period, and the amplitude of image distortion remains consistent. Moreover, when the total integration time of TDI CCD is greater than the jitter period (in this case, it is referred to as high-frequency jitter), which means $N\tau f > 1$. The amplitude of image deformation gradually approaches zero. Therefore, the multistage integration imaging of TDI CCD has a significant impact on the image with high-frequency jitter.

It can be seen from Fig. 4(a) that for the TDI CCD with multistage integration, the amplitude attenuation factor decays with the law of the absolute value of $\sin c$, as shown in the following equation:

$$\alpha = |\sin c(N\tau f)|. \tag{3}$$

B. Relative Jitter Error Detection Using Multispectral Images

A set of optimal band combinations from multispectral images with multistage integration are selected for dense matching to generate the parallax map. The average value of each line in the parallax map was calculated as the relative error of the line. Relative jitter error curves were obtained along the scanning line or during the imaging time. Through Fourier analysis, the frequency, amplitude, and initial phase of the relative jitter error are determined.

The commonly used high-precision and dense matching methods for multispectral images include least-squares image matching [39] and phase correlation matching [29]. The first method uses correlation coefficient matching for coarse registration, followed by least-squares matching. The least-squares matching method is based on image gray values with an initial value correction strategy. It could realize accurate point-by-point dense matching using a small sliding window. The phase correlation matching method is a Fourier-based technique that provides accurate and effective advantages. The sliding window can also be applied to the phase correlation matching method to generally use a larger size. The phase correlation matching method is more accurate for data with low-frequency jitter [40], while the least-squares matching method is more effective for data with high-frequency jitter due to its smaller window size.

C. Rigorous Parallax Observation Model and Inversion of the Absolute Jitter Error With Single-Stage Integration

The previous researches on jitter detection for multispectral images have treated TDI CCD as a single-stage CCD. The

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actual platform jitter parameters were calculated using the parallax observation model through (1).

For multispectral images with multistage integration, the image geometric deformation caused by platform jitter varies among different band images. The registration error between multispectral images is different from that with single-stage integration. Therefore, we proposed a rigorous parallax observation model. It represents the relationship between the relative jitter error of two multispectral images with multistage integration and the absolute jitter error with multistage integration image. And it can be expressed as follows:

$$g_{\mathrm{TDI}}(t) = d_{N_i}(t + \Delta t) - d_{N_j}(t) \tag{4}$$

where t is the current imaging time, and Δt is the imaging time interval. $d_{N_i}(t + \Delta t)$ and $d_{N_j}(t)$ are multispectral image motion with N_i and N_j stages' integration affected by jitter. $g_{\mathrm{TDI}}(t)$ is the relative jitter error between multispectral images with multistage integration.

The platform jitter can be considered as harmonic motion combined with one or more sinusoidal functions [41]. At a short period of time, the image displacement curve affected by jitter can be described by a sinusoidal function, as shown in the following equation:

$$d(t) = A\sin(2\pi f t + \varphi) \tag{5}$$

where A is the amplitude of image motion affected by jitter with single-stage integration, φ is the initial phase, and f is the jitter frequency.

Therefore, the TDI CCD image motion with multistage integration caused by jitter can be expressed as follows:

$$d_N(t) = \alpha A\sin(2\pi f t + \varphi + \Delta\varphi) \tag{6}$$

where α and $\Delta\varphi$ are the amplitude attenuation factor and phase offset of image motion affected by jitter under the imaging time with an integration stage of N, respectively. They are used to detect the actual amplitude and phase of single-stage integration jitter and multistage integration jitter. The difference in TDI CCD integration stage implies that the total integration time varies. The amplitude attenuation factor and phase offset caused by jitter are also not the same.

By substituting (6) into (4), the rigorous parallax observation model can be further expressed as follows:

$$\begin{aligned} g_{\mathrm{TDI}}(t) &= d_{N_i}(t + \Delta t) - d_{N_j}(t) \\ &= \alpha_{N_i} A\sin\left(2\pi f(t + \Delta t) + \varphi + \varphi_{N_i}\right) \\ &\quad - \alpha_{N_j} A\sin\left(2\pi f t + \varphi + \varphi_{N_j}\right) \end{aligned} \tag{7}$$

where $\alpha_{N_i}, \alpha_{N_j}$ and $\varphi_{N_i}, \varphi_{N_j}$ are the amplitude attenuation factors and phase offsets of image motion caused by jitter under the imaging time with an integration stage of N_i and N_j, respectively.

When the registration error between multispectral images is determined, the actual platform jitter with single-stage integration can be deduced through the rigorous parallax observation model. Then, we can model the amplitude and phase of absolute jitter error for TDI CCD image with multistage integration using the amplitude attenuation factor and phase offset through (6).

1) Jitter Amplitude Modeling With Single-Stage Integration of Multistage Integration Images: According to the principles of periodic motion and motion synthesis on the platform, the frequency of the absolute jitter error with single-stage integration should be consistent with the frequency of the relative jitter error.

By combining $g_{\mathrm{TDI}}(t) = A_r\sin(2\pi f t + \varphi_r)$ with (7), we can obtain the relationship in amplitude between the relative jitter error of two multispectral images with multistage integration and the absolute jitter error with single-stage integration, shown as follows:

$$A = \frac{A_r}{\sqrt{\alpha_{N_i}^2 + \alpha_{N_j}^2 - 2\alpha_{N_i}\alpha_{N_j}\cos\left(2\pi f\Delta t + \varphi_{N_i} - \varphi_{N_j}\right)}}. \tag{8}$$

From (8), the amplitude of the relative jitter error is m times greater than that of single-stage integration jitter ($m > 0$). The coefficient m represents the ratio coefficient between the relative jitter error amplitude and the amplitude of the absolute jitter error with single-stage integration, as shown in (9). The ratio coefficient m is not only related to the amplitude attenuation factor, phase offset, integration stages, and single integration time of the two band images but also closely related to the platform jitter frequency and the imaging time interval between the two band images

$$m = \sqrt{\alpha_{N_i}^2 + \alpha_{N_j}^2 - 2\alpha_{N_i}\alpha_{N_j}\cos\left(2\pi f\Delta t + \varphi_{N_i} - \varphi_{N_j}\right)}. \tag{9}$$

Special case 1: when the integration stages of two band images are the same, $\alpha_{N_i} = \alpha_{N_j} = \alpha$ and $N_i = N_j = N$. There are

$$A = \frac{A_r}{\alpha\sqrt{2 - 2\cos(2\pi f\Delta t)}} = \frac{A_r}{2\alpha|\sin(\pi f\Delta t)|}. \tag{10}$$

The sign of A depends on the value of $\pi f\Delta t$. When $\sin(\pi f\Delta t) > 0$, it is positive; conversely, it becomes negative. The value of A is finally positive.

Special case 2: when the two band images are single-stage integration, $\alpha_{N_i} = \alpha_{N_j} = \alpha = 1$ and $N_i = N_j = N = 1$. There are

$$A = \frac{A_r}{2|\sin(\pi f\Delta t)|}. \tag{11}$$

2) Jitter Phase Modeling With Single-Stage Integration of Multistage Integration Images: Let $a = \cos(2\pi f\Delta t + \varphi_{N_i})$, $b = \sin(2\pi f\Delta t + \varphi_{N_i})$, $c = \cos(\varphi_{N_j})$, and $d = \sin(\varphi_{N_j})$, according to the trigonometric function, we know $a^2 + b^2 = 1$ and $c^2 + d^2 = 1$. There are

$$\begin{aligned} g_{\mathrm{TDI}}(t) = A\sqrt{\alpha_{N_i}^2 + \alpha_{N_j}^2 - 2\alpha_{N_i}\alpha_{N_j}\cos\left(2\pi f\Delta t + \varphi_{N_i} - \varphi_{N_j}\right)} \\ \cdot\sin(2\pi f t + \varphi + \theta) \end{aligned} \tag{12}$$

where, $\cos\theta = (a\alpha_{N_i} - c\alpha_{N_j}/((a\alpha_{N_i} - c\alpha_{N_j})^2 + (b\alpha_{N_i} - d\alpha_{N_j})^2)^{1/2})$ and $\sin\theta = (b\alpha_{N_i} - d\alpha_{N_j}/((a\alpha_{N_i} - c\alpha_{N_i})^2 + (b\alpha_{N_i} - d\alpha_{N_j})^2)^{1/2})$.

Therefore, the phase relationship between the relative jitter error of two multispectral images with multistage integration and the absolute jitter error with single-stage integration can be expressed by the following equations:

$$\varphi_r = \begin{cases} \varphi + \theta \\ \pi - (\varphi + \theta) \end{cases} \tag{13}$$

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$$\theta = \arctan\left(\frac{\alpha_{N_i}\cdot\sin\left(2\pi f\Delta t+\varphi_{N_i}\right)-\alpha_{N_j}\cdot\sin\varphi_{N_j}}{\alpha_{N_i}\cdot\cos\left(2\pi f\Delta t+\varphi_{N_i}\right)-\alpha_{N_j}\cdot\cos\varphi_{N_j}}\right). \tag{14}$$

Special case 1: $\alpha_{N_i}=\alpha_{N_j}=\alpha$ and $\varphi_{N_i}=\varphi_{N_j}=\Delta\varphi$, there are

$$\begin{cases} g_{\mathrm{TDI}}(t)=\alpha A\sqrt{2-2\cos(2\pi f\Delta t)}\sin(2\pi f t+\varphi+\Delta\varphi+\theta) \\ \cos\theta=-|\sin(\pi f\Delta t)| \\ \sin\theta=\dfrac{\sin(\pi f\Delta t)\cos(\pi f\Delta t)}{|\sin(\pi f\Delta t)|} \\ \varphi_r=\begin{cases}\varphi+\Delta\varphi+\theta \\ \pi-(\varphi+\Delta\varphi+\theta)\end{cases}\end{cases} \tag{15}$$

where, due to $0<f\Delta t=(\Delta t/T_j)<1, \sin(\pi f\Delta t)>0$, so $\cos\theta=-\sin(\pi f\Delta t)<0$, and $\sin\theta=\cos(\pi f\Delta t)$, there are

$$\theta=\frac{\pi}{2}+\pi f\Delta t. \tag{16}$$

Special case 2: $\alpha_{N_i}=\alpha_{N_j}=\alpha=1$ and $\varphi_{N_i}=\varphi_{N_j}=\Delta\varphi=0$, there are

$$\begin{cases} g_{\mathrm{TDI}}(t)=A\sqrt{2-2\cos(2\pi f\Delta t)}\sin(2\pi f t+\varphi+\theta) \\ \cos\theta=|\sin(\pi f\Delta t)| \\ \sin\theta=\pm\cos(\pi f\Delta t) \\ \varphi_r=\begin{cases}\varphi+\theta \\ \pi-(\varphi+\theta).\end{cases}\end{cases} \tag{17}$$

The value of θ is the same as in special case 1.

D. Jitter Compensation of Panchromatic and Multispectral Images With Multistage Integration

As long as the frequency, amplitude, and phase of the relative jitter error of multispectral images with multistage integration are determined, the absolute jitter error with single-stage integration can be calculated according to (8), (13), and (14). Then, we can obtain the corresponding jitter amplitude attenuation factor and phase offset of panchromatic and multispectral images through their integration stages, single integration time, and jitter frequency, respectively. Absolute jitter error modeling with multistage integration can be, respectively, carried out for panchromatic and multispectral images through (6) based on the single-stage integration jitter error results.

Based on the absolute jitter modeling results of panchromatic and multispectral images, jitter compensation for panchromatic and multispectral images is performed by image resampling to correct geometric distortion of TDI CCD images with multistage integration.

IV. Experiments and Discussion

A. Numerical Simulation Experiments

To verify the correctness of the absolute jitter error inversion with single-stage integration based on the proposed rigorous observation model, the numerical simulation experiment was conducted using TDI CCD multispectral images of rigorous parallax imaging with varying imaging intervals and integration stages.

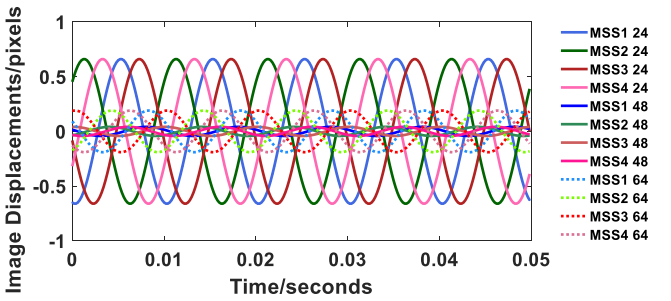

Fig. 5. Simulation results of multispectral images affected by platform jitter under three kinds of integration stages.

According to the design of multispectral images, specific imaging interval between adjacent bands is used to simulate the image displacements for each band. Through (2), we simulated the platform jitter image motion curve of TDI CCD images with different integration stages for each band. We then calculated the relative jitter error between images for different band combinations using (4). The platform jitter parameters (frequency, amplitude, and phase) are estimated using the single-stage integration jitter error inversion model and compared with the input parameters of the simulation to validate the proposed model.

Given the jitter frequency of 100 Hz, the amplitude of 1 pixel, the initial phase of 0 rad, and the single integration time for the TDI camera is 0.0002 s. The integration stages used in the experiments are 24, 48, and 64 TDI stages, respectively. The corresponding amplitude attenuation factors are 0.661307, 0.041625, and 0.190428 pixels, and the phase offsets are −1.759309, 2.752814, and 2.633879 rads. The imaging interval between adjacent bands is 20 pixels, and integration stages of 24, 48, and 64 are used to simulate the image displacements for each band. According to the imaging interval, there are three kinds of band combinations: 1) the combination with an imaging interval of 20 pixels, which includes B2–B1; 2) the combination with an imaging line interval of 40 pixels, including B3–B1; and 3) the combination with an imaging line interval of 60 pixels, including B4–B1. Each band has three kinds of integration stages, and there are six different integration stage combinations within each band combination.

The image displacement curves of different bands under different integration stages are shown in Fig. 5. Under the same integration stage, single integration time, and jitter frequency, the amplitude and frequency of image motion affected by platform jitter are the same in different bands. Due to the difference in imaging interval among bands, their phases are different. There are 18 sets of data, including three kinds of band combinations and six kinds of integration stage combinations. The relative jitter error of each group data is fit with a sine function to obtain the frequency, amplitude, and phase of the relative jitter error. According to the single integration time, the integration stage, the imaging interval, the amplitude attenuation factor, and the phase offset of each

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TABLE I
NUMERICAL SIMULATION RESULTS OF THE FITTING RELATIVE JITTER ERROR AND THE INVERSED SINGLE-STAGE INTEGRATION JITTER ERROR

Combination	Fitting results of relative jitter error			Inversion results of jitter error with single-stage integration			
	Frequency (f_r /Hz)	Amplitude[1] (A_r /pixels)	Phase (φ_r /rad)	m^2	Frequency (f /Hz)	Amplitude[3] (A /pixels)	Phase (φ /rad)
B2-B1:24-24	100.0000	1.257881	1.068092	1.257881	100.0000	1.000000	-0.000032
B2-B1:48-48	100.0000	0.079175	-0.703500	0.079175	100.0000	0.999992	-0.000562
B2-B1:64-64	100.0000	0.362216	-0.821952	0.362216	100.0000	1.000002	-0.000079
B2-B1:24-48	100.0000	0.679622	0.698169	0.679641	100.0000	0.999971	-0.000047
B2-B1:24-64	100.0000	0.741755	0.506865	0.741762	100.0000	0.999990	-0.000039
B2-B1:48-64	100.0000	0.227667	-0.597144	0.227675	100.0000	0.999964	-0.000159
B3-B1:24-24	100.0000	0.777413	2.324729	0.777413	100.0000	1.000000	-0.000032
B3-B1:48-48	100.0000	0.048933	0.553137	0.048933	100.0000	0.999992	-0.000562
B3-B1:64-64	100.0000	0.223862	0.434685	0.223862	100.0000	1.000002	-0.000079
B3-B1:24-48	100.0000	0.625417	-2.983195	0.625405	100.0000	1.000019	-0.000003
B3-B1:24-64	100.0000	0.520177	-2.797563	0.520171	100.0000	1.000013	-0.000021
B3-B1:48-64	100.0000	0.177057	-0.292704	0.177037	100.0000	1.000113	-0.000056
B4-B1:24-24	100.0000	0.777413	0.439773	0.777413	100.0000	1.000000	-0.000032
B4-B1:48-48	100.0000	0.048933	-1.331818	0.048933	100.0000	0.999991	-0.000562
B4-B1:64-64	100.0000	0.223862	-1.450270	0.223862	100.0000	1.000002	-0.000079
B4-B1:24-48	100.0000	0.702680	-0.496003	0.702678	100.0000	1.000003	-0.000064
B4-B1:24-64	100.0000	0.851734	-0.503842	0.851733	100.0000	1.000000	-0.000043
B4-B1:48-64	100.0000	0.186850	-0.728088	0.186869	100.0000	0.999894	-0.000079

[1]: A_r is the amplitude of the relative jitter error between the two different multispectral images.

[2]: m is the ratio coefficient between the relative jitter error amplitude and the amplitude of the absolute jitter error with single-stage integration, $m = \sqrt{\alpha_{N_i}^2 + \alpha_{N_j}^2 - 2\alpha_{N_i}\alpha_{N_j}\cos(2\pi f \Delta t + \varphi_{N_i} - \varphi_{N_j})}$.

[3]: A is the amplitude of the absolute jitter error with single-stage integration, $A = A_r / m$.

group, the inversion results of the single-stage integration jitter error are calculated based on (8), (13), and (14). The results are shown in Table I.

From Table I, it is obvious that the frequency of the relative jitter error f_r is consistent with the frequency of the absolute jitter error with single-stage integration f. The amplitude of the absolute jitter error with single-stage integration A was calculated by formulas $A = A_r/m$, and $m = (\alpha_{N_i}^2 + \alpha_{N_j}^2 - 2\alpha_{N_i}\alpha_{N_j}\cos(2\pi f\,\Delta t + \varphi_{N_i} - \varphi_{N_j}))^{1/2}$ [seen in (8) and (9)]. A_r is the amplitude of the relative jitter error between the two different multispectral images. m is the ratio coefficient between the relative jitter error amplitude and the amplitude of the absolute jitter error with single-stage integration. N_i and N_j are the integration stages of the two multispectral images. α_{N_i} and α_{N_j} are the amplitude attenuation factors of image motion caused by jitter under the imaging time with an integration stage of N_i and N_j, respectively. φ_{N_i} and φ_{N_j} are the phase offsets of image motion caused by jitter under the imaging time with an integration stage of N_i and N_j, respectively. Δt is the imaging interval between the two multispectral images.

To simplify the calculation, we set the amplitude of the absolute jitter error with single-stage integration as 1 pixel. The results presented in Table I indicate that the detected amplitude of the relative jitter error is almost equivalent to the value of the ratio coefficient m. Variations in the set amplitude of the absolute jitter error with single-stage integration will impact the amplitude of relative jitter error, while the ratio coefficient m remains constant under the same imaging parameters (including the single-stage integration time, integration stage, jitter frequency, and imaging interval between the multispectral bands) according to (9). The data in Table I demonstrate that each set of simulation data accurately estimated the frequency, amplitude, and phase of the absolute jitter error with single-stage integration, with an error of no more than 0.0001 pixels. It proved that the rigorous parallax observation model can accurately estimate the absolute jitter error with single-stage integration for multistage integration images, thereby providing the model basis for estimating the absolute jitter error with multistage integration.

B. Real Image Experiments

1) Experimental Data: Based on the jitter error attenuation analysis for TDI CCD image with multistage integration, it was observed that the high-frequency jitter error is significantly influenced by the multistage integration imaging process of TDI CCD. Experiments were conducted using panchromatic and multispectral images obtained from the GaoFen-9 satellite, which suffers the high-frequency jitter. The GaoFen-9 satellite is an optical remote sensing satellite that forms a crucial component of the National High-Resolution Earth Observation system's special project, having been launched on September 14, 2015. The satellite is equipped with a panchromatic and multispectral five-in-one optical TDI camera. The camera focal plane design is shown in Fig. 6.

The optical TDI camera of the GaoFen-9 satellite has five bands arranged in parallel on a focal plane with a certain distance, including the blue band (B1), green band (B2), red band (B3), near-infrared band (B4), and panchromatic band (P) as shown in Fig. 6. These five bands are imaging at the same time t_0 but cover different ground areas due

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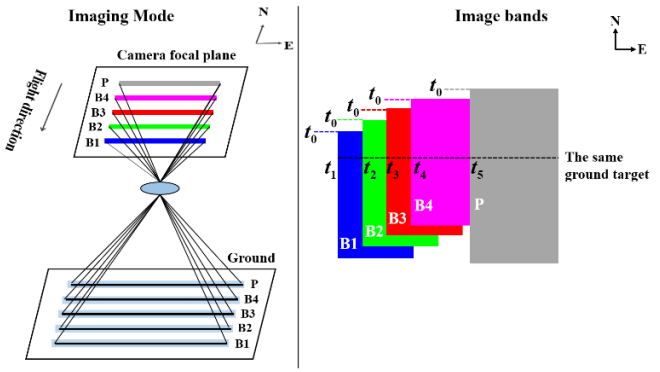

Fig. 6. Schematic of imaging for five-in-one optical TDI camera focal plane.

TABLE II
EXPERIMENTAL DATA OF GAOFEN-9 SATELLITE

Bands	Single integration time (ms)	Integration stages	Size(pixels)
B1	0.362395311	48	701*1536
B2		48	
B3		24	
P	0.0906	48	2801*2500

to the distances between different bands. In other words, different bands capture the same scene at different moments, as illustrated in Fig. 6. Although the ground width of five spectral bands remains consistent, the panchromatic band has higher spatial resolution which is four times greater than that of the multispectral bands. Considering that the radiometric characteristics of infrared band images (B4) are very different from that of visible band images (B1, B2, and B3), images from B1, B2, and B3 bands were used in the experiments. In the TDI camera of the GaoFen-9 satellite, the positions of B1 and B2 bands are 74 pixels apart along the track, while that of B1 and B3 bands are 136 pixels apart. Table II shows the high-resolution image data information of the GaoFen-9 satellite used in the experiment.

For the experimental data, we use the initial imaging time t_0 of B1 image as the reference time. To determine the relative jitter error through image matching between multispectral images, we acquire B2 and B3 images with the same scene as B1 image according to the arrangement interval of the multispectral bands. During the same imaging period as the imaging time of B1, we select the corresponding panchromatic image. The data size of multispectral images is 701 rows by 1536 columns of pixels. The single integration time for multispectral bands is approximately four times longer than that of the panchromatic band. So, the row size of panchromatic image is nearly four times larger than that of multispectral images captured in the same imaging time.

2) Absolute Jitter Error Results With Single-Stage Integration Based on the Proposed Model:

a) Relative jitter error detection results using different band combinations: The dense matching is performed using B2–B1 and B3–B1 band combinations based on B1 band image to obtain parallax images in the across-track and along-track directions. The least-squares matching and phase correlation matching methods were used to detect the relative jitter error. It was observed that the phase correlation matching method was not suitable for the experimental data from the GaoFen-9 satellite due to the high-frequency and low-amplitude jitter present in the satellite images. The larger window size used in the phase correlation matching method compared with the least-squares matching method was unable able to detect the smaller amplitude jitter effectively. Therefore, we used the least-squares matching method to generate the parallax map between different band combinations, as shown in Fig. 7.

To analyze the time-varying characteristics of the relative jitter error, the average value of each line in the parallax image was calculated as the relative jitter error for that line. Subsequently, the relative jitter error curves along the scanning line or over the imaging time were obtained as shown in Figs. 8 and 9. The red boxes in these figures indicate the experimental data used for the subsequent experiments. The relative jitter error curves and the corresponding fitting results are shown in the smaller plots in Figs. 8 and 9. The detection results of relative jitter error are shown in Table III.

Table II shows that B1, B2, and B3 band images have the same single integration time but different integration stages. The degree of jitter distortion varies among B1, B2, and B3 bands, leading to differences in the relative jitter error detected by B2–B1 and B3–B1 bands. In addition, B1 and B2 bands have the same integration stages but have different imaging times at the same scene. So, the absolute jitter error amplitude of B1 and B2 bands is equal, but with different phase. The integration stage of B3 band is smaller than that of B1 and B2 bands, indicating that the jitter attenuation degree in B3 band is lower than in B1 and B2 bands. The absolute jitter error amplitude of B3 band is greater than that of B1 and B2 bands. Therefore, the relative jitter error of B3–B1 bands is larger than that of B2–B1 bands in both the across-track and along-track directions, as shown in Table III. The relative jitter error amplitude of B2–B1 bands is only 0.05 pixels, which is considered very small.

b) Absolute jitter error inversion results with single-stage integration based on the proposed model: In Table IV, the integration stage and single integration time of panchromatic and multispectral images are displayed. We can obtain the amplitude attenuation factor and phase offset based on the value of $N\tau f$, shown in Table IV.

It is noted that a smaller value of the amplitude attenuation factor results in a more pronounced attenuation of the distortion caused by jitter. Consequently, the corresponding absolute jitter error with multistage integration exhibits smaller amplitude compared with that with single-stage integration under the same jitter condition. By using (9), we can calculate the coefficient m for B2–B1 bands and B3–B1 bands according to the corresponding amplitude attenuation factor, phase offset, and imaging interval. The absolute jitter error parameters with single-stage integration under B2–B1 bands and B3–B1 bands can be estimated by applying the absolute jitter error inversion model with single-stage integration [seen in (8), (13), and (14)]. Since B1 and B2 bands have the same imaging parameters, the absolute jitter error parameters with

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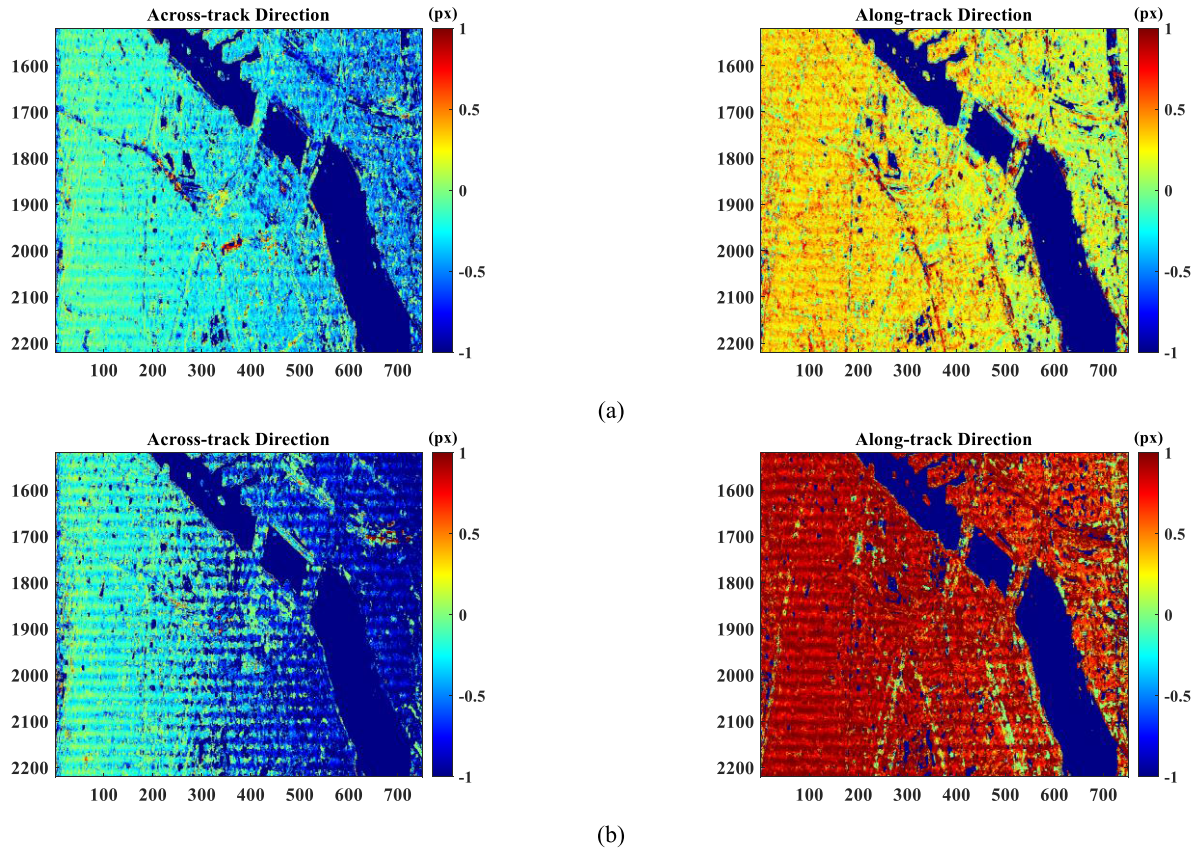

Fig. 7. Obtained parallax images after subpixel matching. (a) Using B2 and B1 bands for matching. (b) Using B3 and B1 bands for matching.

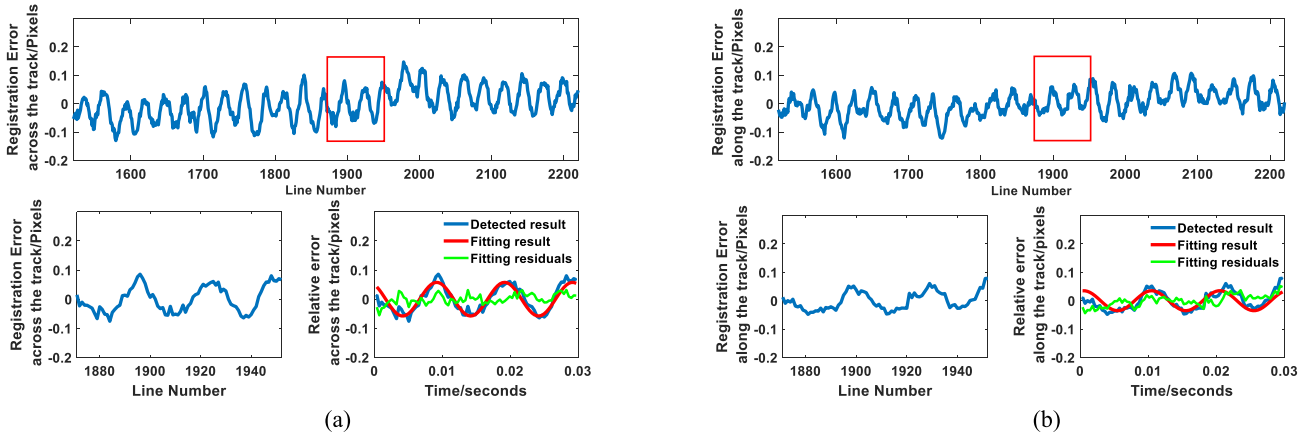

Fig. 8. Relative jitter error curve and fitting results of B2–B1 bands in (a) across-track and (b) along-track directions.

TABLE III
RELATIVE JITTER ERROR DETECTION RESULTS

Combination	Direction	Frequency/Hz	Amplitude/pixels	Phase/rad
B2-B1	Across-track	99.990000	0.057839	2.094062
	Along-track	99.990000	0.035220	1.251559
B3-B1	Across-track	100.0000	0.135155	-0.619875
	Along-track	100.0000	0.061261	-1.516546

single-stage integration under B2–B1 bands can be calculated using (10), (15), and (16). The results are presented in Table V.

The single integration time for panchromatic image is nearly a quarter of that for multispectral images, resulting in the value of $N\tau f$ for panchromatic image being approximately a quarter of that for multispectral images at the same integration stages. Therefore, panchromatic image exhibits a higher amplitude attenuation factor compared with multispectral image.

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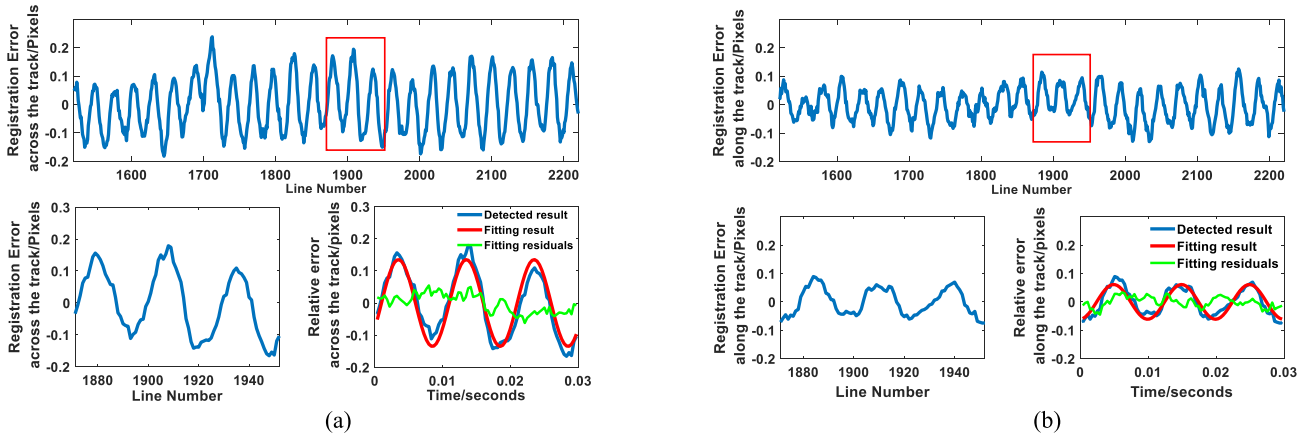

Fig. 9. Relative jitter error curve and fitting results of B3–B1 bands in (a) across-track and (b) along-track direction.

TABLE IV
AMPLITUDE ATTENUATION FACTOR AND PHASE OFFSET OF PANCHROMATIC AND MULTISPECTRAL IMAGES

Band	Single Integration Time(s)	Integration Stages	$N\tau f$	Amplitude attenuation factor	Phase offset
B1	0.000362395311	48	1.739497	0.135111	0.675900
B2		48	1.739497	0.135111	0.675900
B3		24	0.869749	0.146843	1.895067
P	0.0000906	48	0.43488	0.713414	-2.185494

TABLE V
ABSOLUTE JITTER ERROR RESULTS WITH SINGLE-STAGE INTEGRATION

Band combination	m[1]	Direction	Frequency/Hz	Amplitude/pixels	Phase/rad
B2-B1	0.223075	Across-track	99.990000	0.259281	3.960090
					2.913559
		Along-track	99.990000	0.157885	3.117588
					3.756062
B3-B1	0.130672	Across-track	100.000000	1.034304	-3.267088
					1.114254
		Along-track	100.000000	0.468816	-4.163759
					2.010925

[1]: m is the ratio coefficient between the relative jitter error amplitude and the amplitude of the absolute jitter error with single-stage integration. $m = \sqrt{\alpha_{N_i}^2 + \alpha_{N_j}^2 - 2\alpha_{N_i}\alpha_{N_j}\cos(2\pi f \Delta t + \varphi_{N_i} - \varphi_{N_j})}$.

The relative jitter error results for B2–B1 and B3–B1 bands are different, and their coefficient m is also different. However, the jitter error inversion results with single-stage integration based on B2–B1 and B3–B1 bands are not consistent. Therefore, the absolute jitter error modeling and compensation with multistage integration are carried out to compare the accuracy of the absolute jitter error inversion results with single-stage integration.

3) Jitter Compensation of Panchromatic and Multispectral Images With Multistage Integration: Based on the results of Tables IV and V, the absolute jitter error modeling with multistage integration for panchromatic and multispectral images is carried out in the across-track and along-track directions, as described in (6).

The jitter error modeling results with multistage integration based on B2–B1 and B3–B1 bands using the proposed model and traditional model are shown in Tables VI and VII, respectively. To establish a consistent time reference, the initial imaging time t_0 of each band image is used as reference time in real image experiments. Subsequent modeling of relative jitter error and absolute jitter error for five bands is carried out based on the same reference time. The traditional jitter detection model does not account for the impact of multistage integration imaging for TDI CCD on high-frequency jitter. The initial phases of all the bands are the same, which is equal to the estimated initial phase of the absolute jitter error. Similarly, the amplitudes of multispectral images are also the same. According to the spatial resolution relationship between panchromatic and multispectral images, the amplitude of the absolute jitter error in the panchromatic image is four times greater than that in the multispectral image. The initial phase of the panchromatic image is the same as that of the multispectral images.

However, the amplitude and phase of absolute jitter error using B2–B1 and B3–B1 bands differ in the proposed model in our article, while the frequency remains the same. In addition, the amplitude and phase results of absolute jitter error for panchromatic and multispectral images in the traditional model

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TABLE VI
ABSOLUTE JITTER ERROR MODELING RESULTS OF PANCHROMATIC AND MULTISPECTRAL IMAGES BASED ON B2–B1 BANDS

		The modeling results based on traditional jitter detection model			The modeling results based on the presented model		
		Frequency/Hz	Amplitude/pixels	Phase/rads	Frequency/Hz	Amplitude/pixels	Phase/rads
Across-track	B1	99.990000	0.035032	1.494398	99.990000	0.035032	7.777583
	B2	99.990000	0.035032	1.494398	99.990000	0.035032	3.589460
	B3	99.990000	0.035032	1.494398	99.990000	0.038074	4.808627
	P	99.990000	0.140128	1.494398	99.990000	0.184975	3.869658
Along-track	B1	99.990000	0.021332	0.651895	99.990000	0.021332	7.573555
	B2	99.990000	0.021332	0.651895	99.990000	0.021332	6.935080
	B3	99.990000	0.021332	0.651895	99.990000	0.023184	8.154247
	P	99.990000	0.085328	0.651895	99.990000	0.112637	4.712161

TABLE VII
ABSOLUTE JITTER ERROR MODELING RESULTS OF PANCHROMATIC AND MULTISPECTRAL IMAGES BASED ON B3–B1 BANDS

		The modeling results based on traditional jitter detection model			The modeling results based on the presented model		
		Frequency/Hz	Amplitude/pixels	Phase/rads	Frequency/Hz	Amplitude/pixels	Phase/rads
Across-track	B1	100.000000	0.523876	1.080277	100.000000	0.139746	0.550404
	B2	100.000000	0.523876	1.080277	100.000000	0.139746	1.790154
	B3	100.000000	0.523876	1.080277	100.000000	0.151880	3.009321
	P	100.000000	2.095504	1.080277	100.000000	0.737887	2.070353
Along-track	B1	100.000000	0.237456	0.183606	100.000000	0.063342	5.828418
	B2	100.000000	0.237456	0.183606	100.000000	0.063342	-0.346267
	B3	100.000000	0.237456	0.183606	100.000000	0.068842	0.872900
	P	100.000000	0.949824	0.183606	100.000000	0.334460	2.967024

are also different from those in our model. To further compare and analyze the accuracy of the inversion results, we proceeded with experiments on distortion correction of panchromatic and multispectral images.

Based on the absolute jitter error modeling results of panchromatic and multispectral images, the bilinear resampling method is used for jitter compensation and to obtain the corrected panchromatic and B1, B2, and B3 band images. The distance between the bands used was then eliminated through translation, and the true color multispectral image was synthesized for display. The compensation performance for multispectral image can be compared by displaying the true color image before and after jitter distortion correction. In instances where notable jitter is present in the panchromatic image, the compensation effectiveness can be evaluated by isolating the edges of linear characteristics from the panchromatic image both before and after jitter distortion correction.

a) Jitter compensation results of multispectral images: The compensation results of the experiments are illustrated in Fig. 10. The original image details are depicted in Fig. 10(a), while Fig. 10(b) and (c) exhibits the jitter compensation results obtained through the traditional jitter detection model using B2–B1 and B3–B1 bands. On the other hand, Fig. 10(d) and (e) displays the jitter compensation results achieved by the proposed method using B2–B1 and B3–B1 bands.

It is obvious that a visible registration error exists in the original images, whereas the jitter distortion in the resultant images has been improved to varying extents. In terms of visual impact, Fig. 10(e) demonstrates that the proposed method using B3–B1 bands offers the best compensation performance and effectively eliminates the registration error. Due to the same integration stages of B1 and B2 bands, Fig. 10(b) and (d) shows similar compensation performance, both failing to correct the jitter distortion due to the significant extent of jitter attenuation. Fig. 10(c) indicates that the compensation results by the traditional model using B3–B1 bands cannot effectively restore the true jitter error of multispectral images, resulting in increased geometric distortion due to the lack of consideration for multistage integration.

Furthermore, quantitative comparisons were conducted to evaluate performances, with the evaluation indices for the band-to-band registration results of multispectral images before and after jitter compensation shown in Tables VIII and IX. These evaluation indices include average error (AE), root mean square error (RMSE), maximum error (Max. E), and minimum error (Min. E). Analysis of Table VIII reveals that the registration results based on B2–B1 and B3–B1 bands, derived from multispectral images restored by the proposed model using the B2–B1 band combination, exhibit similar values to those obtained from the traditional model. The registration errors calculated by both the methods closely align with the results prior to jitter compensation, indicating that the utilization of B2–B1 band combination in the two methods does not effectively correct jitter distortion. Conversely, Table IX shows that the evaluation indices of the registration error for B2–B1 and B3–B1 bands, calculated by the proposed model using B3–B1 band combination, are smaller than those obtained by the traditional model, particularly in the cross-track direction. Notably, the registration errors after jitter distortion correction via the proposed method are significantly reduced compared with those prior to distortion correction.

b) Jitter compensation results of panchromatic image: The original panchromatic image is depicted in Fig. 11(a)–(c),

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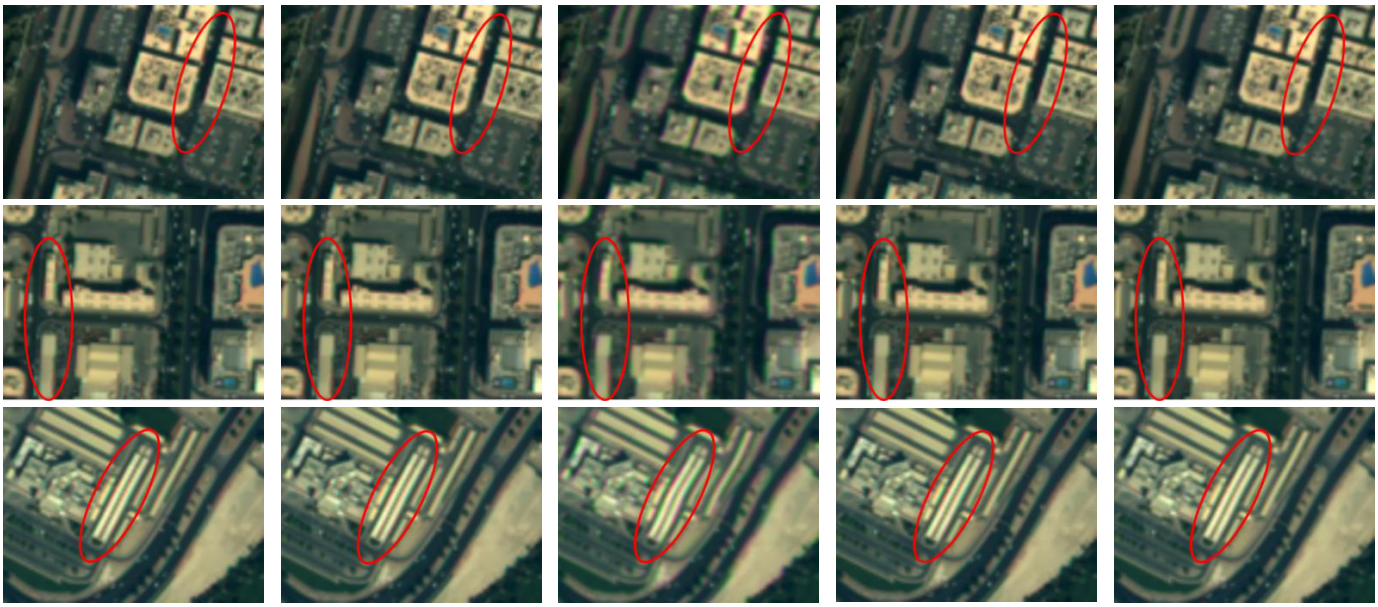

(a) (b) (c) (d) (e)

Fig. 10. Jitter compensation results on multispectral image of the GaoFen-9 satellite. (a) Details of original multispectral image. (b) and (c) Jitter compensation results of multispectral image by the traditional jitter detection model using B2–B1 and B3–B1 band combinations. (d) and (e) Jitter compensation results of multispectral image by the proposed method using B2–B1 and B3–B1 band combinations.

TABLE VIII

BAND-TO-BAND REGISTRATION ERROR BEFORE AND AFTER JITTER COMPENSATION FOR MULTISPECTRAL IMAGES USING B2–B1 BANDS (UNIT: PIXELS)

Band Combination	Direction	Original image				Compensated image by the traditional method				Compensated image by the proposed method			
		AE	RMSE	Max. E	Min. E	AE	RMSE	Max. E	Min. E	AE	RMSE	Max. E	Min. E
B2-B1	Across-track	-0.0057	0.0770	0.1157	-0.1275	-0.0078	0.0544	0.0784	-0.0954	-0.0060	0.0798	0.1163	-0.1288
	Along-track	0.0229	0.0598	0.1150	-0.0727	0.0214	0.0471	0.0924	-0.0546	0.0200	0.0406	0.0812	-0.0376
B3-B1	Across-track	-0.0410	0.1678	0.2358	-0.2843	-0.0409	0.1685	0.2371	-0.2827	-0.0416	0.1699	0.2455	-0.2871
	Along-track	-0.0577	0.1084	0.0826	-0.1964	-0.0577	0.1095	0.0846	-0.1956	-0.0559	0.1024	0.0730	-0.1839

TABLE IX

BAND-TO-BAND REGISTRATION ERROR BEFORE AND AFTER JITTER COMPENSATION FOR MULTISPECTRAL IMAGES USING B3–B1 BANDS (UNIT: PIXELS)

Band Combination	Direction	Original image				Compensated image by the traditional method				Compensated image by the proposed method			
		AE	RMSE	Max. E	Min. E	AE	RMSE	Max. E	Min. E	AE	RMSE	Max. E	Min. E
B2-B1	Across-track	-0.0057	0.0771	0.1157	-0.1275	-0.0524	0.4358	0.5753	-0.6299	-0.0095	**0.0522**	**0.0722**	**-0.0785**
	Along-track	0.0229	0.0598	0.1150	-0.0727	0.0063	0.1859	0.3221	-0.2554	0.0220	0.0803	0.1391	-0.0930
B3-B1	Across-track	-0.0409	0.1678	0.2358	-0.2843	-0.0482	0.1815	0.2476	-0.3192	-0.0360	**0.0588**	**0.0529**	**-0.1169**
	Along-track	-0.0577	0.1084	0.0826	-0.1964	-0.0463	0.1023	0.0914	-0.1999	-0.0428	0.0801	0.0845	-0.1444

with the edge detection results of linear features in the red region illustrated by the green dotted line in Fig. 11(c) and further detailed in Fig. 11(d). Figs. 12 and 13 show the compensation and edge detection results of panchromatic image in the across-track direction using the traditional jitter detection model with B2–B1 and B3–B1 bands. In addition, Figs. 14 and 15 show the compensation and edge detection results of panchromatic image by the proposed method using B2–B1 and B3–B1 bands, respectively.

It is obvious that the compensation results achieved by the proposed method using B3–B1 bands show better compensation performance, and Fig. 15 shows that the edge detection result is almost a straight line. Likewise, the jitter compensation performance depicted in Figs. 12 and 14 does

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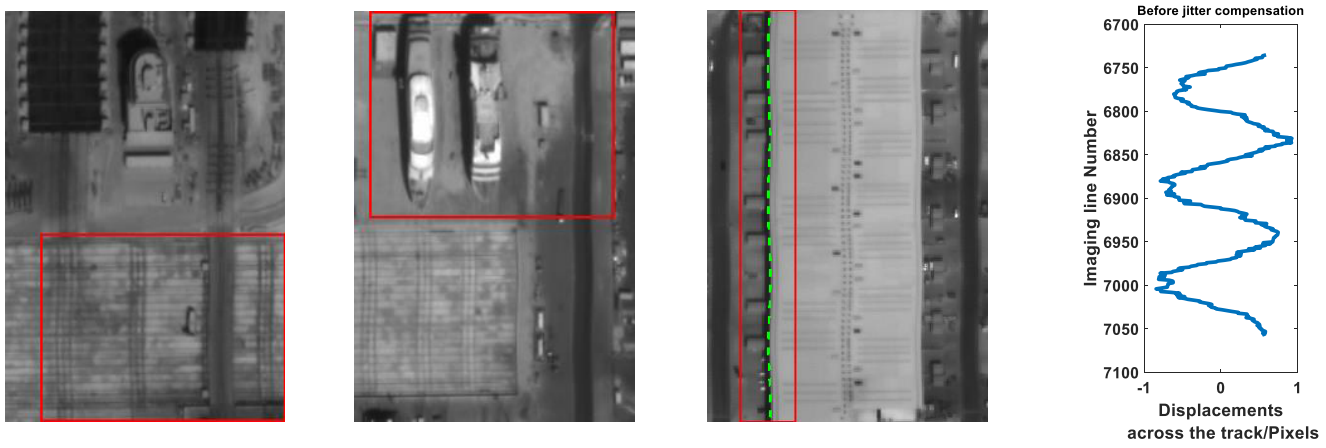

Fig. 11. Original panchromatic image. (a)–(c) Details of original panchromatic image. (d) Edge detection result of linear feature in (c).

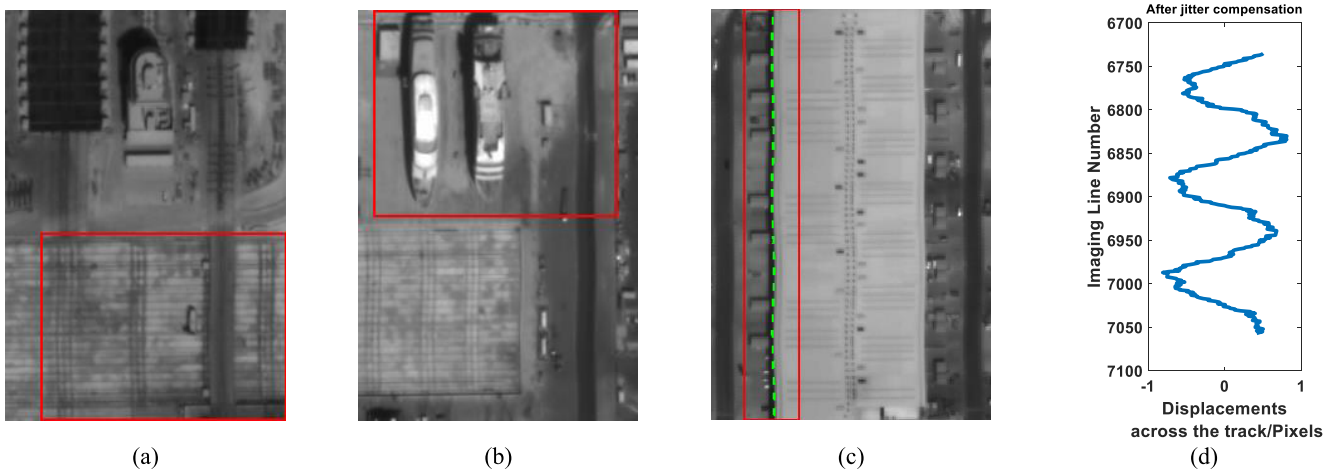

Fig. 12. Result of panchromatic image after jitter compensation by the traditional jitter detection model using B2–B1 band combination. (a)–(c) Details of panchromatic image. (d) Edge detection result of linear feature in (c).

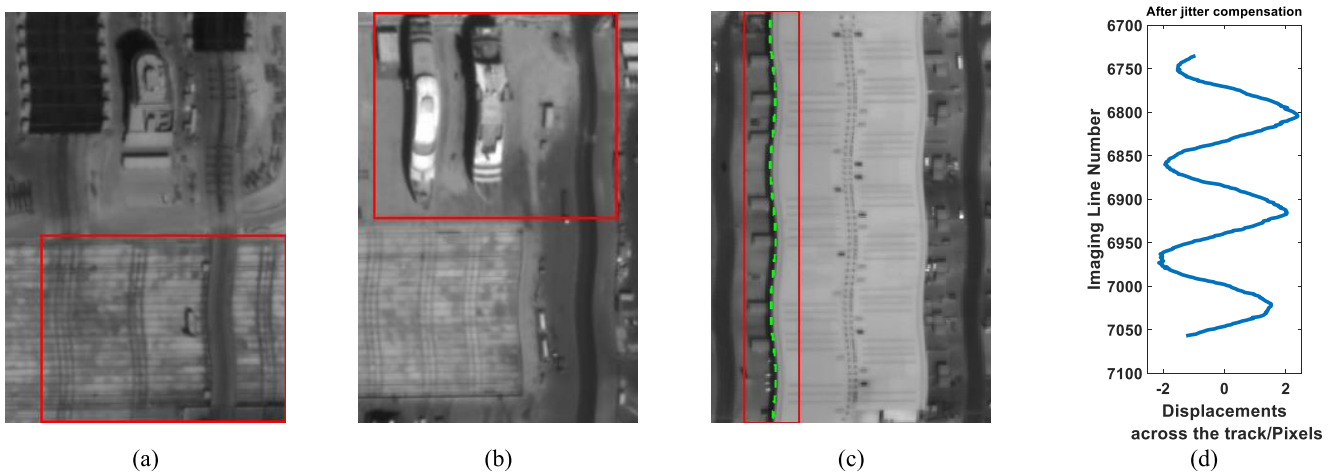

Fig. 13. Result of panchromatic image after jitter compensation by the traditional jitter detection model using B3–B1 band combination. (a)–(c) Details of panchromatic image. (d) Edge detection result of linear feature in (c).

not show significant. However, the jitter distortion correction result in Fig. 13 indicates that the traditional jitter detection model using B3–B1 bands is unable to eliminate the jitter error and may even exacerbate the jitter distortion.

4) Discussion:

a) Analysis of jitter error modeling and compensation results based on the traditional model and the proposed model: According to the jitter error modeling and the jitter

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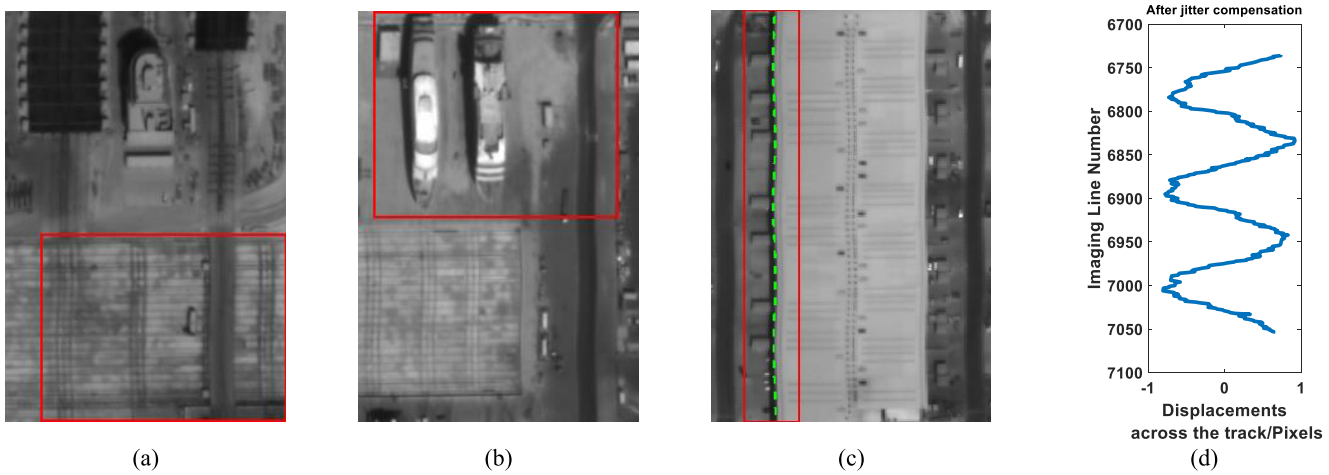

Fig. 14. Result of panchromatic image after jitter compensation by the proposed model using B2–B1 band combination. (a)–(c) Details of panchromatic image. (d) Edge detection result of linear feature in (c).

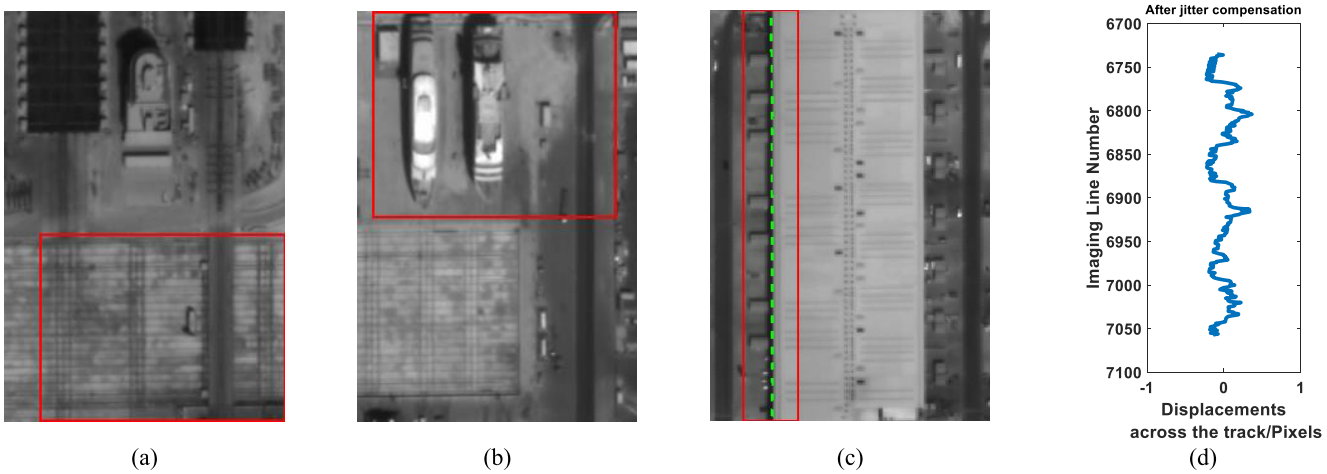

Fig. 15. Result of panchromatic image after jitter compensation by the proposed model using B3–B1 band combination. (a)–(c) Details of panchromatic image. (d) Edge detection result of linear feature in (c).

distortion correction results, neither B2–B1 band combination nor B3–B1 band combination can accurately detect the absolute jitter error of each band using the traditional jitter detection model. The traditional model failed to effectively eliminate the jitter error and even exacerbated the jitter distortion when using B3–B1 band combination.

The modeling results for B1 and B2 bands shown in Table VI by the two methods using B2–B1 band combination are approximately the same. However, Table VII shows the jitter modeling results for panchromatic and multispectral images by the two methods using B3–B1 band combination are different. The presented method can detect the absolute jitter error with multistage integration based on the absolute jitter error with single-stage integartion. The amplitude of the absolute jitter error with single-stage integration A is calculated by formulas $A = A_r/m$, and $m = (\alpha_{N_i}^2 + \alpha_{N_j}^2 - 2\alpha_{N_i}\alpha_{N_j}\cos(2\pi f\Delta t + \varphi_{N_i} - \varphi_{N_j}))^{1/2}$ (which is mentioned in Sections III-B and IV-A). Then, the absolute jitter error for panchromatic and multispectral images by the proposed model is calculated by formula $d_N(t) = \alpha A\sin(2\pi f t + \varphi + \Delta\varphi)$ (α is the amplitude attenuation factors of the multistage integration image with an integration stage of N). Therefore, the amplitude of the absolute jitter error for multistage integration images can be estimated by formula $A_N = \alpha A$. The traditional jitter detection model ignores the multistage integration imaging process of TDI CCD. It does not involve the absolute jitter error detection with single-stage or multistage integration. It only establishes the relationship between the relative jitter error and the absolute jitter error, and the amplitude of the absolute jitter error can be calculated by formula $A_d = (A_r/2|\sin(\pi f\Delta t)|)$ (A_d is the amplitude of absolute jitter error in the traditional model).

According to the above formulas, the calculated value of α/m is smaller than that of $1/2|\sin(\pi f\Delta t)|$. Therefore, the amplitudes of absolute jitter error for multistage integration images in the presented model are also smaller compared with those in the traditional model shown in Table VII. The initial phase of the absolute jitter error with multistage integration is similar to the calculation of the amplitude. So, for multistage integration images with high-frequency jitter, the absolute jitter error modeling results using the proposed method differ from those obtained by the traditional method, unless the

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integration stage of the two multispectral images used in image matching are the same. In such instances, the absolute jitter error modeling results for the two multispectral images used in image matching are consistent between both the methods, as seen in Table VI.

For B2–B1 band combination, the integration stage, single integration time, and jitter frequency are the same for both B1 and B2 bands. The amplitude of absolute jitter error with single-stage integration of the B2–B1 band can be determined through the formula $A = (A_r/2\alpha|\sin(\pi f \Delta t)|)$ [in (10)]. Consequently, the calculation of the absolute jitter error amplitude for B1 and B2 bands becomes $A_{B_1} = A_{B_2} = \alpha_{B_1} \cdot A = (A_r/2|\sin(\pi f \Delta t)|)$, which means that it degrades into the traditional jitter detection model. As a result, the calculated absolute jitter error amplitudes for B1 and B2 band images using the traditional model and the proposed model in this article, based on B2–B1 bands, are the same in the across-track and along-track directions but exhibit different phases.

However, B3 band image has a different integration stage compared with B1 and B2 band images. Due to the consideration with multistage integration processing of TDI CCD in our model, the absolute jitter error amplitude calculated based on the proposed model differs from that in the traditional model for B3 band image and panchromatic image (as shown in Table VI). As for B2–B1 band combination, the absolute jitter error modeling results based on both the traditional model and the proposed model failed to recover the true jitter amplitude and phase information. These results did not show any significant improvement on jitter compensation for panchromatic and multispectral images, shown in Figs. 10(b) and (d), 12, and 14. B1 and B2 band images have the larger integration stage, and B1 and B2 band images experienced greater jitter attenuation. So, the absolute jitter error of B1 and B2 band images is not obvious, which means that the absolute jitter error with single-stage integration using B2–B1 bands is hard to be estimated. The absolute jitter error with multistage integration cannot be accurately modeled.

For B3–B1 band combination, B1 and B3 band images are at different integration stages. The absolute jitter modeling results based on the traditional model and the proposed model are not the same (as shown in Table VII). The experimental data have high frequency of 100 Hz, which suffered severe jitter attenuation. The traditional model did not consider the TDI integration stages, and the true jitter errors were not correctly estimated for panchromatic and multispectral images (as shown in Figs. 10(c) and 13). From Table VII, the modeling results show the amplitude and phase of absolute jitter error in the across-track and along-track directions for panchromatic image. Specifically, the traditional model using B3–B1 band combination yielded values of 2.095504 pixels and 1.080277 rads for across-track amplitude and phase, and 0.949824 pixels and 0.183606 rads for along-track amplitude and phase, respectively. In contrast, the proposed model using the same band combination had values of 0.737887 pixels and 2.070353 rads for across-track amplitude and phase, and 0.334460 pixels and 2.967027 rads for along-track amplitude and phase. During the jitter distortion using the traditional method, discrepancies in phase estimation resulted in misalignment between the modeled and actual absolute jitter error curves. Moreover, inaccuracies in larger amplitude estimation led to excessive correction during jitter compensation, exacerbating image deformation. Similar correction performance was observed in the jitter distortion correction experiments conducted on multispectral images using B3–B1 bands within the traditional model. The proposed model which considers the impact caused by the TDI integration stage can accurately recover the true jitter amplitude and phase information (as shown in Table VII) and effectively correct the jitter distortion for panchromatic and multispectral images (as shown in Figs. 10(e) and 15). The experimental results show that the proposed method achieved better results than the traditional model for data with high frequency.

According to the jitter error attenuation analysis for TDI CCD images with multistage integration, multistage integration images suffer slight jitter attenuation under low-frequency condition. This means that the TDI CCD can be treated as a single-stage integration CCD in this case, especially when the amplitude attenuation factors are all close to 1 pixel, and the phase offsets are all near 0 rad. The presented model degrades into the traditional model (seen in the special case 2 in Section III-C), indicating that the absolute jitter error modeling results and jitter compensation performance between the proposed method and the traditional method are very close. For instance, the jitter frequency is 1 Hz, and the other imaging parameters match the experimental data in this article. The amplitude attenuation factors of B1, B2, B3, and P bands are 0.999443, 0.999443, 0.999849, and 0.999943 pixels, respectively. The corresponding phase offsets are 0.165758, 0.165758, 0.082482, and 0.046792 rads. The jitter distortion correction performances in the traditional model and the proposed model are similar. Consequently, the traditional method may not be effective for TDI CCD images with high-frequency jitter, but it still performs well on data with low-frequency jitter. Compared with the traditional model, the proposed rigorous parallax observation model has wider applicability.

b) Analysis of jitter modeling results and compensation performance between different band combinations: In our real image experiments, the number of integration stages for B1, B2, and B3 band images is 48, 48, and 24 TDI stages, respectively. According to the jitter attenuation law for TDI CCD images with multistage integration, under the same conditions of jitter frequency and single integration time, the extent of jitter attenuation is greater with a larger number of integration stages. Therefore, B1 and B2 bands have the same degree of jitter attenuation. The jitter attenuation degree in the B3 band is smaller than that in the B1 and B2 bands. Thus, the B3 band exhibits a larger amplitude of absolute jitter error compared with the B1 and B2 bands (as shown in Tables VI and VII). It can be found that the jitter error in B1 and B2 band images attenuates to a small value. The detection accuracy of relative jitter error is lower when using B2–B1 band combination compared with B3–B1 band combination. This results in the inability to estimate the true jitter error with single-stage integration and correct jitter distortion for multistage integration images.

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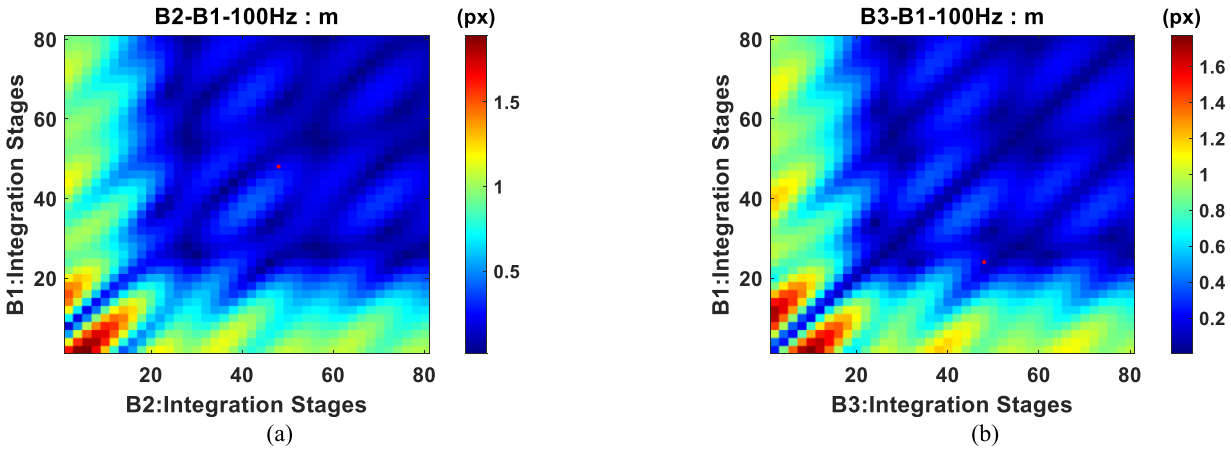

Fig. 16. Map of the ratio coefficient m between the amplitude of the relative jitter error and the absolute jitter amplitude with single-stage integration under different imaging time intervals. (a) B2–B1 band combination. (b) B3–B1 band combination.

According to (8), the amplitude of absolute jitter error with single-stage integration for multispectral images is related to the coefficient m. For further analysis, we calculate the ratio coefficient m using different imaging time intervals and integration stages through (9). The single integration time, jitter frequency, and imaging interval between different bands are the same as in the real image experiments.

The ratio coefficient maps for B2–B1 and B3–B1 band combinations are shown in Fig. 16(a) and (b), respectively. The difference in the ratio coefficient m is solely related to the imaging time interval between different band combinations. It is obvious that the values of the coefficient m for B2–B1 and B3–B1 band combinations are distributed in the interval of (0, 2). When the integration stage of both the bands is larger than 20, the ratio coefficient value of m for B3–B1 band combination is mainly smaller than that for B2–B1 band combination. Both are primarily distributed in the interval of (0, 0.5). The accuracy of amplitude in jitter error with single-stage integration depends on the precision of detecting relative jitter error once the ratio coefficient m is determined. The detection accuracy of relative jitter error with multistage integration requires higher precision. Moreover, the detection error of absolute jitter with single integration is smaller when ratio coefficient m has a larger value.

We also display the ratio coefficient m when the integration stage of B1, B2, and B3 bands is 48, 48, and 24 TDI stages, respectively, as shown in the red dots of Fig. 16. The ratio coefficient m for B3–B1 band combination is smaller than that for B2–B1 bands (as shown in Table V). In this case, the difference in the ratio coefficient m is not only related to the imaging time interval between bands but also to the amplitude attenuation factor and phase offset of different bands. Table III shows that the relative jitter error of B3–B1 band combination is significantly larger than that of B2–B1 band combination. So, the proposed method can accurately estimate the actual jitter error parameters using B3–B1 band combination.

Therefore, the accuracy of jitter modeling with multistage integration, based on the method proposed in this article, is closely related to the degree of jitter error attenuation in the two band images. When the jitter error of the both band images attenuate to a small extent, resulting in a smaller relative jitter error, the actual jitter error with single-stage integration cannot be accurately estimated. This results in a lack of precision in jitter modeling with multistage integration, as well as the inability to correct jitter distortion for multistage integration images.

V. Conclusion

In this article, we present a rigorous parallax observation model-based remote sensing panchromatic and multispectral images' jitter distortion correction method for TDI cameras. The main innovation of this model lies in the introducing of jitter amplitude attenuation factor and phase offset, as well as the absolute jitter error inversion model with single-stage integration for multispectral images based on rigorous parallax observation model. Traditional jitter detection and compensation methods typically treat the TDI CCD as a single-stage integration CCD or use blind restoration methods to restore degraded images. When dealing with data with high-frequency jitter, the TDI effect can lead to significant jitter attenuation, rendering the traditional jitter detection method ineffective. So, our method considers the integration stages of TDI CCD, analyzes the jitter attenuation law, and detects jitter from the degradation mechanism of TDI CCD images. We can obtain the jitter amplitude attenuation factor and phase offset under different integration stages and integration times. Moreover, our method achieves the detection from the relative jitter error with multistage integration to the absolute jitter error with single-stage integration, and then to the absolute jitter error with multistage integration. Panchromatic and multispectral images of the GaoFen-9 satellite are adopted in real image jitter distortion correction experiments. Quantitative evaluation and comparison results with the traditional method show that our approach offers advantages in jitter detection and compensation with multistage integration for high-frequency data.

Nevertheless, the subpixel matching based on the least-squares matching algorithm used in this article for high-frequency jitter detection may have its limitations. Other, more accurate matching methods may further increase the

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accuracy of parallax images, thereby enhancing the performance of relative jitter detection. In addition, the multistage integration images for TDI CCD have an image blurring problem, which could affect the accuracy of jitter detection with multistage integration. These issues will be addressed in the future.

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